Class 11: AlphaFold

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```r
library(bio3d)

pdb <- read.pdb("HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_

pdb
```

```
 Call:  read.pdb(file = "HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrela

   Total Models#: 1
     Total Atoms#: 1514,  XYZs#: 4542  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 0  (residues: 0)
     Non-protein/nucleic resid values: [ none ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, calpha, call
```

```r
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y      z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 16.922 -3.898 -6.254 1 90.81
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 16.891 -2.467 -6.562 1 90.81
3 ATOM     3     C <NA>   PRO     A     1   <NA> 16.406 -1.617 -5.395 1 90.81
4 ATOM     4    CB <NA>   PRO     A     1   <NA> 15.930 -2.373 -7.746 1 90.81
5 ATOM     5     O <NA>   PRO     A     1   <NA> 15.820 -2.146 -4.445 1 90.81
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 15.031 -3.559 -7.598 1 90.81
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     C   <NA>
5  <NA>     O   <NA>
6  <NA>     C   <NA>
```

Make a vector of input PDB file names that we can read into R

```r
pdbfiles <- list.files("HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/", pattern=".pdb", full.na

pdbfiles
```

```
[1] "HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_001_alphafol
[2] "HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_002_alphafol
[3] "HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_003_alphafol
[4] "HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_004_alphafol
[5] "HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_005_alphafol
```

Below is the molstar image overlying it. Unfortunately it is not possible to do both parts of the dimer for overlaying since there are some strange errors, so I just aligned the right side.

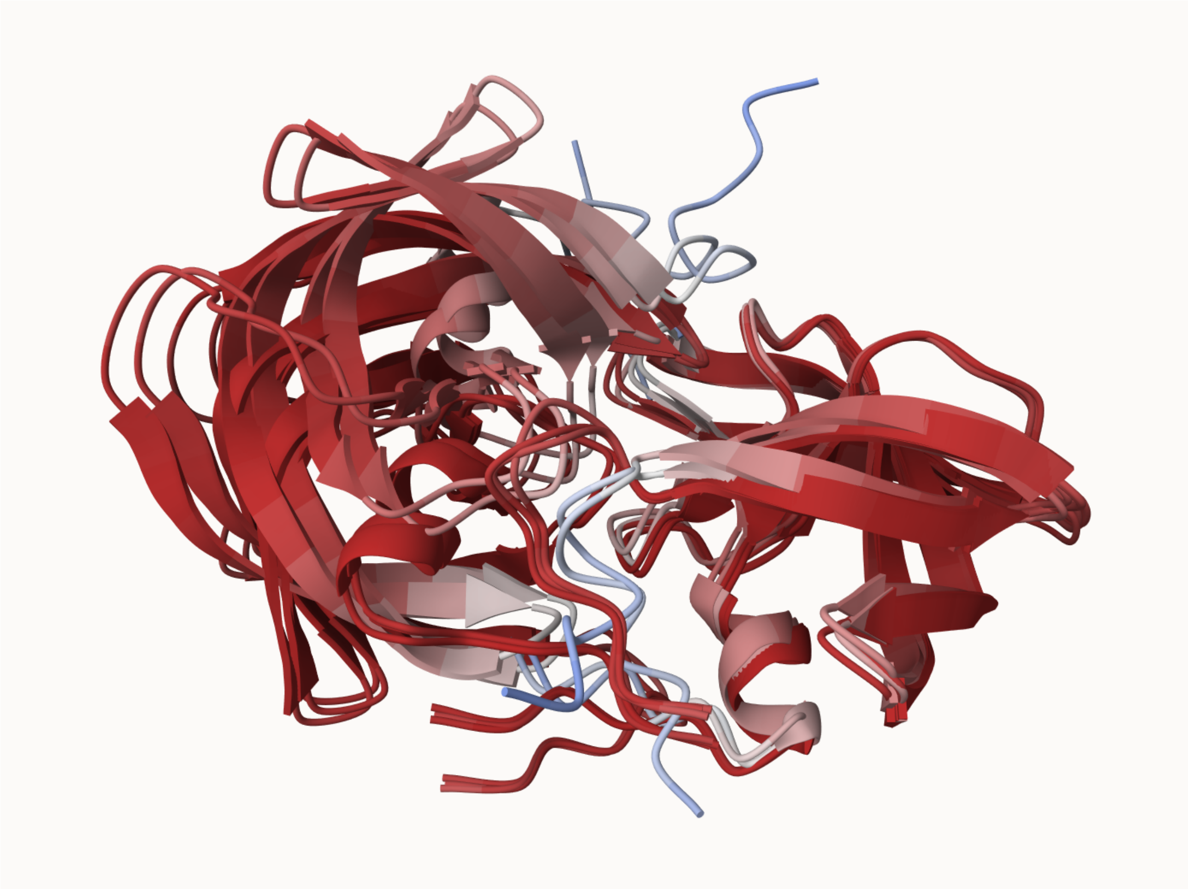


Figure 1: HIV Dimer Structure Overlay

Custom analysis of resulting models

```
pdbs <- pdbaln(pdbfiles, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_001_alphafold2_mu
HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_002_alphafold2_mu
HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_003_alphafold2_mu
HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_004_alphafold2_mu
HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_rank_005_alphafold2_mu
.....

Extracting sequences
```

```
pdb/seq: 1   name: HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_ra
pdb/seq: 2   name: HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_ra
pdb/seq: 3   name: HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_ra
pdb/seq: 4   name: HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_ra
pdb/seq: 5   name: HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0_unrelaxed_ra
```

```
pdbs
```

```
                                1         .         .         .         .         50
[Truncated_Name:1]HIV_DIMER_    PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]HIV_DIMER_    PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]HIV_DIMER_    PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]HIV_DIMER_    PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]HIV_DIMER_    PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                                **************************************************
                                1         .         .         .         .         50

                               51         .         .         .         .         100
[Truncated_Name:1]HIV_DIMER_    GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]HIV_DIMER_    GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]HIV_DIMER_    GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]HIV_DIMER_    GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]HIV_DIMER_    GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                                **************************************************
                               51         .         .         .         .         100

                              101         .         .         .         .         150
[Truncated_Name:1]HIV_DIMER_    QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]HIV_DIMER_    QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]HIV_DIMER_    QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]HIV_DIMER_    QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]HIV_DIMER_    QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                                **************************************************
                              101         .         .         .         .         150

                              151         .         .         .         .      198
[Truncated_Name:1]HIV_DIMER_    GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]HIV_DIMER_    GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]HIV_DIMER_    GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]HIV_DIMER_    GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]HIV_DIMER_    GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
```

```
                                   ************************************************
                                 151         .         .         .         .       198

Call:
  pdbaln(files = pdbfiles, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```r
rd <- rmsd(pdbs, fit=T)
```

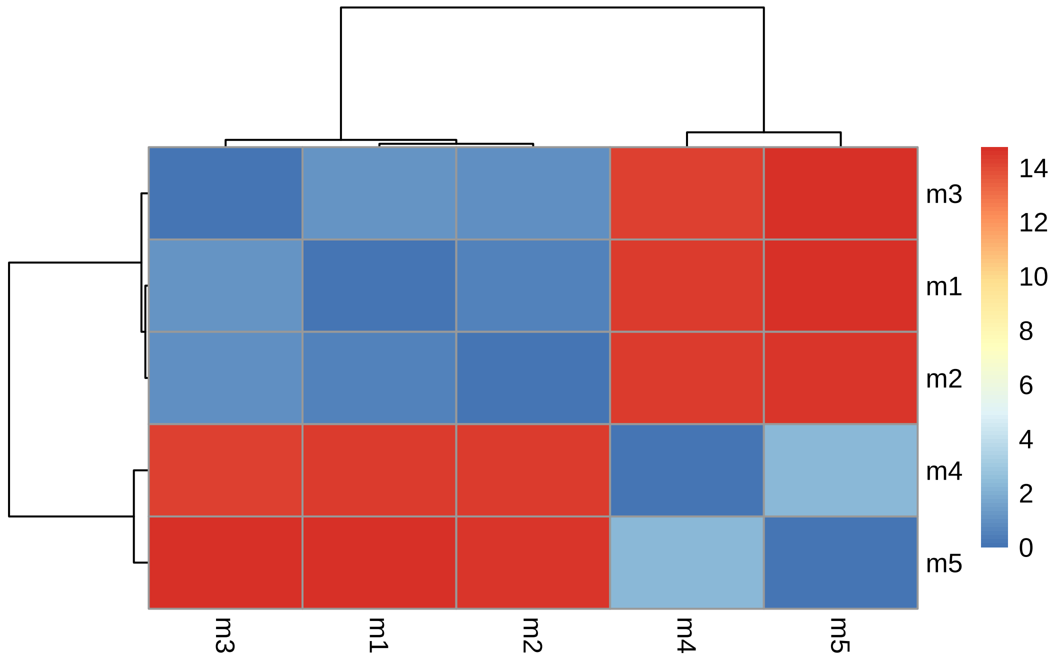
```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.754
```

Drawing a heatmap of the RMSD matrix values

```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



Below will be a read of all the pLDDT values

```r
pdb2 <- read.pdb("1HSG")
```

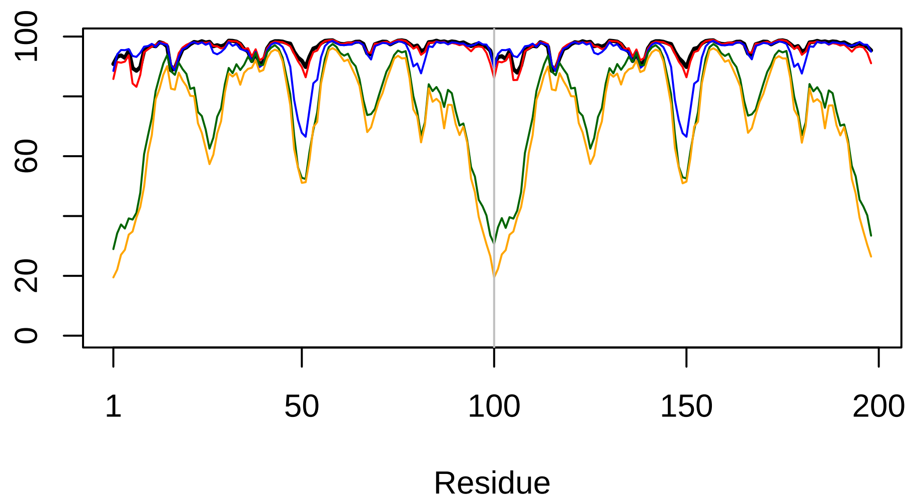
```
  Note: Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
```

```
Warning in pdb2sse(sse): No helix and sheet defined in input 'sse' PDB object:
try using dssp()
```

```
Warning in plotb3(pdbs$b[1, ], typ = "l", lwd = 2, sse = pdb): Length of input
'sse' does not equal the length of input 'x'; Ignoring 'sse'
```

```r
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 9881.716
core size 196 of 198  vol = 6878.296
core size 195 of 198  vol = 1334.229
core size 194 of 198  vol = 1039.217
core size 193 of 198  vol = 950.298
core size 192 of 198  vol = 897.651
core size 191 of 198  vol = 833.382
core size 190 of 198  vol = 769.99
core size 189 of 198  vol = 731.732
core size 188 of 198  vol = 695.939
core size 187 of 198  vol = 658.409
core size 186 of 198  vol = 623.889
core size 185 of 198  vol = 588.148
core size 184 of 198  vol = 566.834
core size 183 of 198  vol = 543.661
core size 182 of 198  vol = 511.572
core size 181 of 198  vol = 489.388
core size 180 of 198  vol = 468.918
core size 179 of 198  vol = 449.633
core size 178 of 198  vol = 433.621
```

```
core size 177 of 198  vol = 419.234
core size 176 of 198  vol = 405.688
core size 175 of 198  vol = 392.453
core size 174 of 198  vol = 381.499
core size 173 of 198  vol = 371.883
core size 172 of 198  vol = 356.017
core size 171 of 198  vol = 345.622
core size 170 of 198  vol = 336.503
core size 169 of 198  vol = 325.735
core size 168 of 198  vol = 314.055
core size 167 of 198  vol = 303.257
core size 166 of 198  vol = 293.703
core size 165 of 198  vol = 284.865
core size 164 of 198  vol = 278.254
core size 163 of 198  vol = 266.143
core size 162 of 198  vol = 258.428
core size 161 of 198  vol = 247.147
core size 160 of 198  vol = 239.31
core size 159 of 198  vol = 234.436
core size 158 of 198  vol = 229.567
core size 157 of 198  vol = 221.505
core size 156 of 198  vol = 215.181
core size 155 of 198  vol = 206.319
core size 154 of 198  vol = 197.35
core size 153 of 198  vol = 188.035
core size 152 of 198  vol = 181.788
core size 151 of 198  vol = 176.501
core size 150 of 198  vol = 170.249
core size 149 of 198  vol = 165.704
core size 148 of 198  vol = 159.379
core size 147 of 198  vol = 153.34
core size 146 of 198  vol = 148.708
core size 145 of 198  vol = 143.325
core size 144 of 198  vol = 136.797
core size 143 of 198  vol = 132.187
core size 142 of 198  vol = 126.89
core size 141 of 198  vol = 121.226
core size 140 of 198  vol = 116.432
core size 139 of 198  vol = 112.209
core size 138 of 198  vol = 107.817
core size 137 of 198  vol = 104.77
core size 136 of 198  vol = 100.894
core size 135 of 198  vol = 97.097
```

```
core size 134 of 198  vol = 92.711
core size 133 of 198  vol = 87.976
core size 132 of 198  vol = 83.829
core size 131 of 198  vol = 81.711
core size 130 of 198  vol = 77.856
core size 129 of 198  vol = 75.599
core size 128 of 198  vol = 72.912
core size 127 of 198  vol = 70.604
core size 126 of 198  vol = 68.832
core size 125 of 198  vol = 66.515
core size 124 of 198  vol = 64.216
core size 123 of 198  vol = 60.981
core size 122 of 198  vol = 58.864
core size 121 of 198  vol = 56.456
core size 120 of 198  vol = 53.855
core size 119 of 198  vol = 51.645
core size 118 of 198  vol = 49.503
core size 117 of 198  vol = 48.055
core size 116 of 198  vol = 46.501
core size 115 of 198  vol = 44.58
core size 114 of 198  vol = 43.121
core size 113 of 198  vol = 40.925
core size 112 of 198  vol = 38.993
core size 111 of 198  vol = 36.323
core size 110 of 198  vol = 33.973
core size 109 of 198  vol = 31.337
core size 108 of 198  vol = 29.321
core size 107 of 198  vol = 27.222
core size 106 of 198  vol = 25.732
core size 105 of 198  vol = 24.059
core size 104 of 198  vol = 22.571
core size 103 of 198  vol = 20.997
core size 102 of 198  vol = 19.912
core size 101 of 198  vol = 18.312
core size 100 of 198  vol = 16.852
core size 99 of 198  vol = 14.362
core size 98 of 198  vol = 13.67
core size 97 of 198  vol = 10.548
core size 96 of 198  vol = 8.367
core size 95 of 198  vol = 6.437
core size 94 of 198  vol = 4.976
core size 93 of 198  vol = 4.06
core size 92 of 198  vol = 3.676
```

```
 core size 91 of 198  vol = 2.77
 core size 90 of 198  vol = 2.155
 core size 89 of 198  vol = 1.712
 core size 88 of 198  vol = 1.144
 core size 87 of 198  vol = 0.866
 core size 86 of 198  vol = 0.683
 core size 85 of 198  vol = 0.528
 core size 84 of 198  vol = 0.37
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

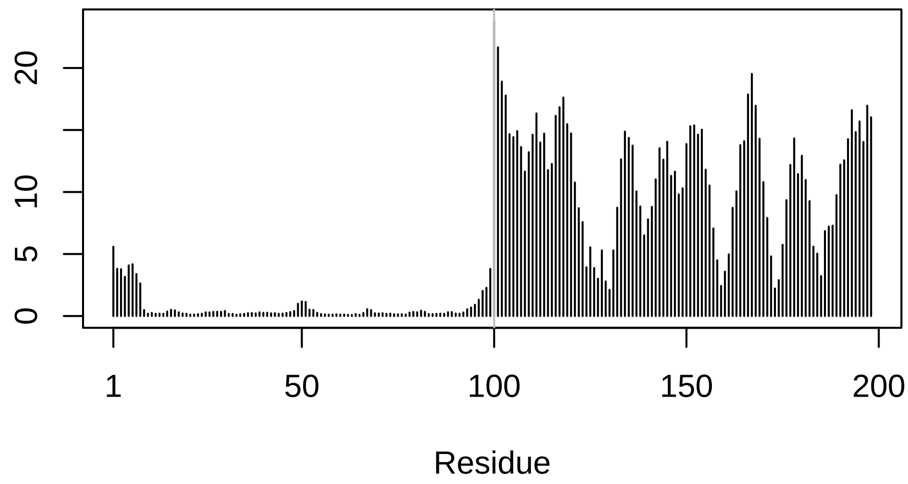
```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)



plotb3(rf, sse=pdb)
```

```
Warning in pdb2sse(sse): No helix and sheet defined in input 'sse' PDB object:
try using dssp()
```

```
Warning in plotb3(rf, sse = pdb): Length of input 'sse' does not equal the
length of input 'x'; Ignoring 'sse'
```

```r
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

```
library(jsonlite)

results_dir <- "HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/"
# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

Maximum PAE values of each model to determine which has the most error:

```r
pae1$max_pae
```

```
[1] 12.84375
```

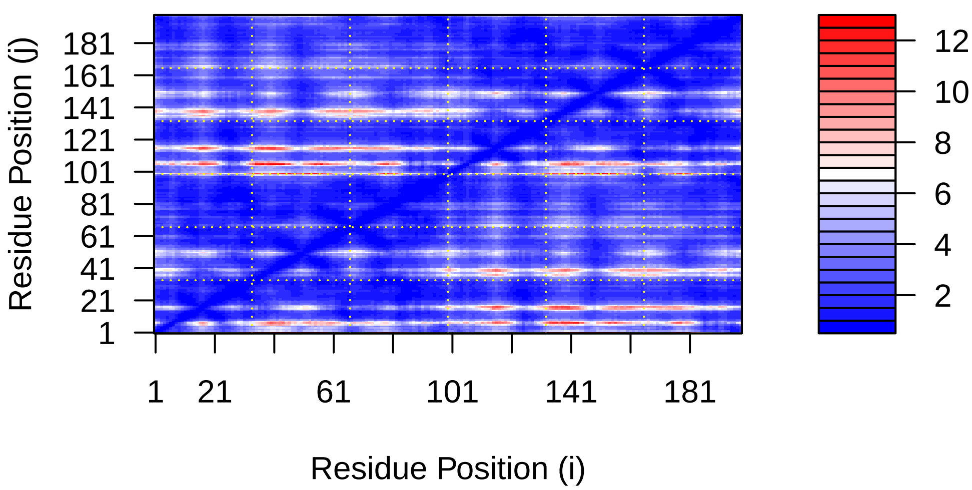
```r
pae5$max_pae
```

```
[1] 29.59375
```

Model 1 has lower PAE than model 5 thus is better due to having lower error.

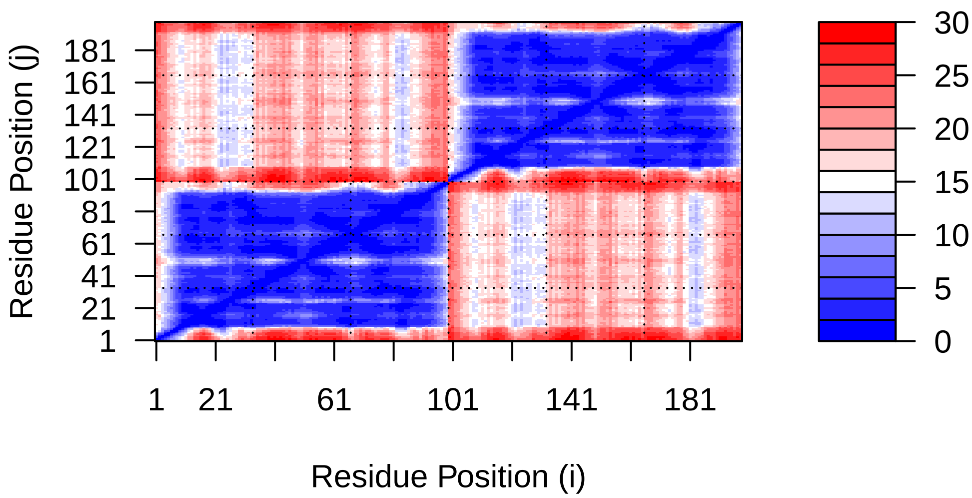
The below is an overall error frame showing the residues in the prediction vs the real one for model 1:

```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```



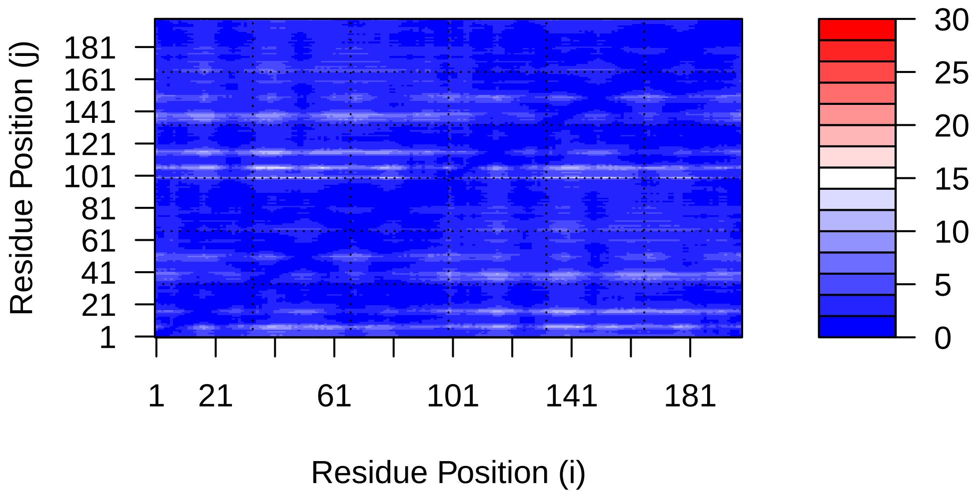
Whereas the following is from model 5:

```
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



But the ranges are messed up because model 1 shows more finer error thus use the following to actually view it:

```
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue Conservation from alignment file

Loads the file a3m into a variable for easier use:

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "HIV_DIMER_23119_0.result/HIV_DIMER_23119_0/HIV_DIMER_23119_0.a3m"
```

Reads the fasta from the file above into a data frame:

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

Checks the number of sequences in this alignment:

```r
dim(aln$ali)
```

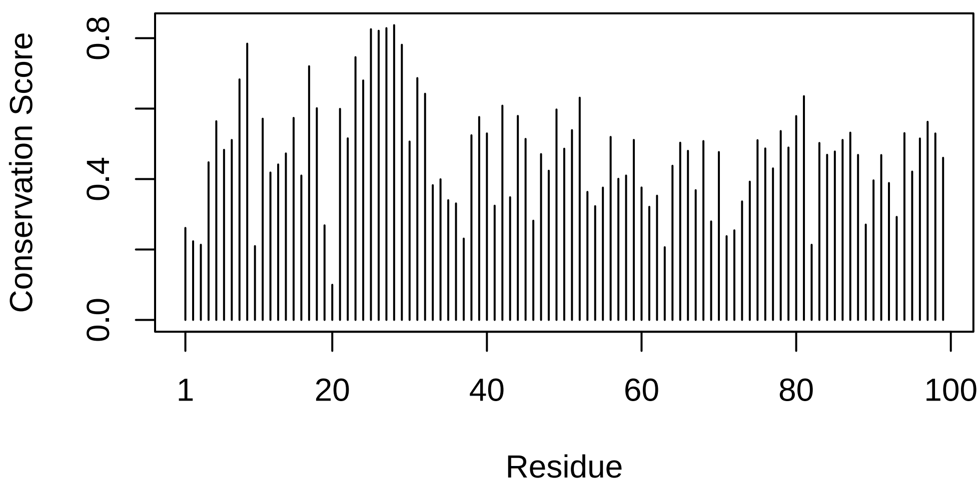
```
[1] 5397  132
```

The below ranks sequences by their conservation amount for certain residues.

```r
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```

```
Warning in pdb2sse(sse): No helix and sheet defined in input 'sse' PDB object:
try using dssp()
```

```
Warning in plotb3(sim[1:99], sse = trim.pdb(pdb, chain = "A"), ylab =
"Conservation Score"): Length of input 'sse' does not equal the length of input
'x'; Ignoring 'sse'
```




```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

Visualization of important sites:

The below shows a consensus structure based on a consensus sequence. This was received from molstar.

```r
m1.pdb <- read.pdb(pdbfiles[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

The below shows purple as more conserved sites and white as more variable sites

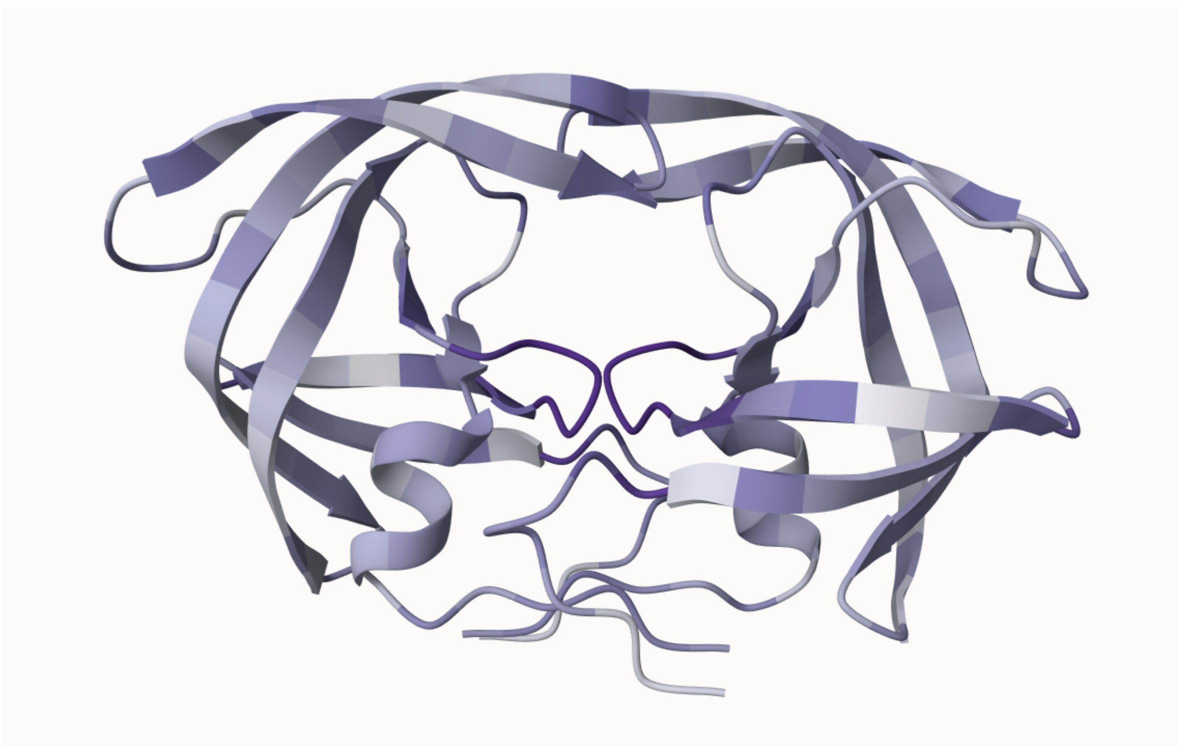


Figure 2: 1HSG Predicted Consensus Structure